Indian Institute of Technology Jodhpur

Department of Mathematics

MAL6150: Probability and Statistics
Quiz 2 — Questions and Solutions

Date: 24 October 2026 **Maximum marks:** $10 + 2$ (bonus)

---

1. **[3 marks]**

Suppose X is a discrete random variable with probability mass function

$$\mathbb{P}(X = n) = c\,4^{-n}, \qquad n = 1, 2, 3, \ldots,$$

where $c > 0$ is a constant.

(a) Determine c. [1 mark]

(b) Compute $\mathbb{E}[X]$. [2 marks]

Solution. Normalisation gives

$$1 = c\sum_{n=1}^{\infty} 4^{-n} = \frac{c}{3}, \qquad \text{so} \qquad \boxed{c = 3}.$$

Differentiating the geometric series yields $\sum_{n\geq 1} nx^n = x/(1-x)^2$ for $|x| < 1$. Therefore

$$\boxed{\mathbb{E}[X] = 3\sum_{n=1}^{\infty} \frac{n}{4^n} = 3\frac{1/4}{(1-1/4)^2} = \frac{4}{3}}.$$

2. **[3 marks]**

Suppose $X \geq 0$ and

$$\mathbb{P}(X > t) = \frac{1}{(1+t)^{3/2}}, \qquad t \geq 0.$$

Determine all real numbers $r > 0$ for which $\mathbb{E}[X^r]$ is finite.

Solution. By the tail-integral formula,

$$\mathbb{E}[X^r] = r\int_0^{\infty} \frac{t^{r-1}}{(1+t)^{3/2}}\,dt.$$

For $0 < t < 1$, since $1 < 1 + t < 2$,

$$2^{-3/2}t^{r-1} \leq \frac{t^{r-1}}{(1+t)^{3/2}} \leq t^{r-1}.$$

Thus the integral over $(0, 1)$ converges for every $r > 0$. For $t \geq 1$, the inequalities $t \leq 1 + t \leq 2t$ give

$$2^{-3/2}t^{r-5/2} \leq \frac{t^{r-1}}{(1+t)^{3/2}} \leq t^{r-5/2}.$$

By comparison with $\int_1^{\infty} t^{r-5/2}\,dt$, the second part converges exactly when $r - 5/2 < -1$. Hence

$$\boxed{\mathbb{E}[X^r] < \infty \iff 0 < r < \frac{3}{2}}.$$

3. **[4 marks]**

Let X be uniformly distributed on $(-1, 1)$, and put $Y = X^2$.

(a) Find the distribution function $F_Y(y)$ and the density $f_Y(y)$, specifying their values for all real y.

(b) Sketch the graphs of $f_X(x)$ and $f_Y(y)$, clearly indicating their supports.

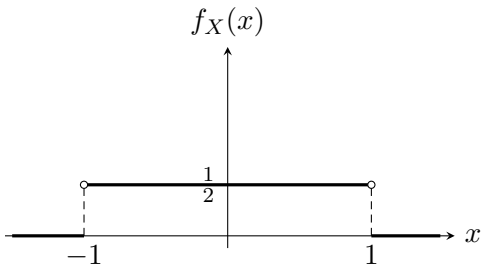
Solution. For $0 \le y < 1$,

$$F_Y(y) = \mathbb{P}(X^2 \le y) = \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) = \sqrt{y}.$$

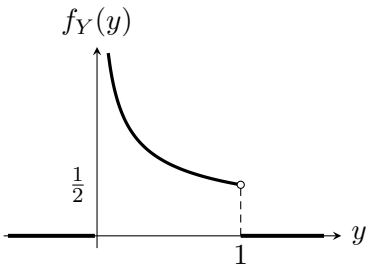
Consequently,

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \sqrt{y}, & 0 \le y < 1, \\ 1, & y \ge 1, \end{cases} \qquad f_Y(y) = \begin{cases} \dfrac{1}{2\sqrt{y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

The density $f_X(x)$ is $1/2$ for $-1 < x < 1$ and zero otherwise. The density f_Y decreases from $+\infty$ near 0 to $1/2$ as $y \uparrow 1$, and vanishes outside $(0, 1)$.



Density of X



Density of Y

Bonus question **[2 marks]**

Two fair six-sided dice are rolled independently. Let M denote the maximum of the two outcomes. Compute $\mathbb{E}[M]$.

Solution. The following three approaches are all valid.

Method 1: Directly find the mass function. For $k = 1, \dots, 6$, exactly $k^2 - (k-1)^2 = 2k - 1$ ordered outcomes have maximum k. Hence

$$\mathbb{P}(M = k) = \frac{2k-1}{36}, \qquad \mathbb{E}[M] = \frac{1}{36}\sum_{k=1}^{6} k(2k-1) = \frac{2(91) - 21}{36} = \boxed{\frac{161}{36}}.$$

Method 2: Use the absolute difference. Writing X_1, X_2 for the dice outcomes,

$$M = \frac{X_1 + X_2 + |X_1 - X_2|}{2}, \qquad \mathbb{E}[|X_1 - X_2|] = \frac{2}{36}\sum_{d=1}^{5} d(6-d) = \frac{70}{36}.$$

As $\mathbb{E}[X_1] = \mathbb{E}[X_2] = 7/2$,

$$\mathbb{E}[M] = \frac{7}{2} + \frac{1}{2} \cdot \frac{70}{36} = \boxed{\frac{161}{36}}.$$

Method 3: Use the tail-sum identity. Independence gives $\mathbb{P}(M \le k) = (k/6)^2$, so

$$\mathbb{E}[M] = \sum_{k=1}^{6} \mathbb{P}(M \ge k) = \sum_{k=1}^{6} \left(1 - \frac{(k-1)^2}{36}\right) = 6 - \frac{55}{36} = \boxed{\frac{161}{36}}.$$